

AquaGuard AI

Smart Facility & Sustainability Manager · Commercial Airport Operations

THE COMMERCIAL FACILITY CHALLENGE

- **Unnoticed Water Wastage:** A single leaking flush valve loses 2.7 L/min (3,888 L/day = 116,640 L/month = ■5,657/month).
- **Blind Maintenance:** High-traffic airport restrooms rely on periodic visual patrols, catching leaks hours or days after occurrence.
- **Phantom Flushes & Wear:** Unsynchronized solenoids and pressure surges cause ghost flushes and premature component fatigue.
- **Manual Hygiene Monitoring:** Scheduled cleaning ignores volatile flight arrival surges, leading to service complaints and passenger hygiene breaches.

THE AQUAGUARD AI AUTONOMOUS SOLUTION

- **Sub-Second Detection:** Continuous flow and phantom flush rules detect anomalies within 3 seconds of onset.
- **Physics-Based Sensor Fusion:** Correlates flow, IR occupancy, acoustic cavitation (2,420 Hz), and traveler QR feedback.
- **Automated Computer-Aided Dispatch:** Automatically cuts work orders, reserves Kohler OEM spares, and enforces 15-min SLAs.
- **Closed-Loop Verification:** Live telemetry mathematically verifies flow reduction to 0.00 L/min before banking accredited savings.

CORE TRANSFORMATION: IoT Telemetry → AI Anomaly Intelligence → Water Waste Detection → Predictive Maintenance → Automated Dispatch → Verified Conservation Ledger

"From reactive manual maintenance to autonomous, research-grade facility intelligence."

End-to-End System Architecture

Modular, decoupled industrial intelligence stack for commercial aviation infrastructure

1. INGESTION & TWIN • 97 Smart Fixtures (Tripoint) • 12 Restrooms · 4 Terminals • Flow (L/min) · Pressure (bar) • Optical IR Occupancy • Hydrophone Cavitation (Hz) • MQTT / Telemetry API Gateway	2. DETECTION & ML • Diurnal Baseline Subtraction • Deterministic Continuous Flow • Solenoid Phantom Cycle Rules • Multimodal Sensor Fusion • Bayesian Leak Confidence • Weibull Degradation Hazard	3. INCIDENT INTEL • Canonical Incident Schema • P1/P2/P3 SLA Dynamic Timer • Kohler OEM Spare Matching • Depot Shelf Inventory Sync • CAD Technician Dispatch • Maximo / SAP PM Work Orders	4. VERIFY & LEDGER • Real-Time Flow Decay Probe • Zero-Leak Baseline Audit • Fixture Health Restoration • Scope 2/3 GHG Carbon Nexus • LEED v4.1 Platinum Scorecard • SHA-256 Digital Certificate
GROUNDING AI COMMAND CENTER (REACT REASONING LAYER) Natural Language Operations Engine operating via strict function-calling whitelist: <code>get_water_waste_heatmap()</code>, <code>get_facility_health_hierarchy()</code>, <code>get_device_detail()</code>, <code>simulate_inaction()</code>. Zero Generative Hallucination: All metrics, tariffs (■48.50/kL), and volumes are computed by deterministic engines before being explained.			

Live Demonstration & Technical Innovations

Autonomous closed-loop incident lifecycle verified on Pune Airport Terminal 2 · **Video:**
<https://drive.google.com/file/d/1LHxz6cDNqUZJG2bPqfCbQhXoR5tncc0D/view?usp=sharing>

STEP 1: INJECT LEAK • Simulate continuous leak on FV-182 (Restroom 14, Stall 03). • Flow jumps to 2.70 L/min. • Zero traveler occupancy. • Line pressure steady at 3.0 bar.	STEP 2: DETECTION & CAD • Anomaly score spikes to 94/100. • Acoustic hydrophone detects 2,420 Hz screech (14.5 Hz flutter). • CAD Schematics: EPDM Diaphragm pulses red in exploded CAD view. • Loss flagged: 116,640 L/month.	STEP 3: 1-CLICK DISPATCH • Priority P1 Critical Alert. • 15-minute SLA timer countdown. • OEM Spare Requisitioned: KOHLER-GP1138930 (T2 Rack B-04). • CMMS Work Order WO-2026-00182 generated with LOTO safety protocol.	STEP 4: VERIFY & BANK • Technician replaces diaphragm. • Verification probe confirms flow drops back to 0.00 L/min. • Fixture health restored: 98/100. • 116,640 Liters saved banked to official sustainability ledger.
Acoustic Cavitation Hydrophone Web Audio API physical frequency synthesis (420 Hz laminar, 2,420 Hz cavitation tear, 120 Hz solenoid hum) with real-time canvas oscilloscope.	Exploded CAD Schematics Interactive vector architectural diagram of Kohler K-10673 flushometer with dynamic drop-shadow fault highlighting and OEM parts registry.	National Portfolio Fleet Multi-airport portfolio switcher across Pune PNQ (Live), Mumbai BOM T2, and Delhi DEL T3 aggregating 593 fixtures and 260.9k L/day.	

Measurable Enterprise Impact & Value

Quantified operational, environmental, and financial returns for commercial facility managers

WATER STEWARDSHIP • 116,640 L/month conserved per resolved critical leak. • 260,900 L/day saved across 3 monitored airports. • ■3,522,150/month in commercial utility tariff costs avoided (@ ■48.50/kL). • 11.9 commercial water tankers eliminated per single incident.	OPERATIONAL EXCELLENCE • Automated P1 Dispatch: Technician dispatched within 3 seconds of leak confirmation. • 15-Minute SLA Guarantee: Real-time countdown prevents regulatory breach. • Zero Manual Logging: Automated IBM Maximo / SAP PM work order generation. • 100% Closed-Loop Verification: Eliminates false maintenance sign-offs.
PREDICTIVE ASSET HEALTH • Weibull Hazard Modeling: Flags failure risk before physical rupture occurs. • Vibration Chatter Index: Solves water hammer and premature coil burnout. • Zero Flight Bank Downtime: Proactive spares staging in T2 Depot Shelf B-04. • Hardware Payback: Sensor retrofit amortized in 4.8 months.	ESG & GREEN BUILDING • Scope 2/3 GHG Avoidance: 212.7 kg CO■e offset monthly (1.82 kWh/kL CEA nexus). • LEED v4.1 Platinum Tier: 92/100 points achieved on water efficiency credits. • Cryptographic Assurance: Tamper-proof SHA-256 digital certificate seals. • Adaptive Hygiene Threshold: Automatic janitorial dispatch on flight deplaning.

DETECT · DIAGNOSE · PRIORITIZE · DISPATCH · CONSERVE